

HOUSE PRICE PREDICTION – PROJECT SUMMARY

Introduction:

This project aims to predict house prices using machine learning techniques and identify the key factors that influence property values. The Housing Dataset contains information about various house characteristics such as area, bedrooms, bathrooms, stories, parking, and furnishing status.

Data Preprocessing:

The dataset was cleaned by checking for missing values and duplicate records. Categorical features were converted into numerical format using one-hot encoding to make the data suitable for machine learning algorithms. Relevant features were selected for building the prediction models.

Model Development:

Two machine learning models were implemented: Linear Regression and Random Forest Regressor. The dataset was divided into training and testing sets, and both models were trained to predict house prices based on the selected features.

Model Evaluation:

The models were evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score. Linear Regression achieved an MAE of 932,640.44, RMSE of 1,251,802.93, and R^2 Score of 0.5503. Random Forest Regressor achieved an MAE of 936,060.35, RMSE of 1,264,548.31, and R^2 Score of 0.5411. Based on these metrics, Linear Regression performed slightly better than Random Forest Regressor.

Data Visualization and Insights:

Several visualizations were created, including a histogram of house prices, a correlation heatmap, and an actual versus predicted price scatter plot. The analysis showed that area, number of bathrooms, stories, and air conditioning had the strongest influence on house prices. The histogram indicated that most houses fall within the low to medium price range, while only a few properties have very high prices.

Key Findings:

The study revealed that larger houses with more amenities generally have higher prices. Area was identified as the most important feature affecting house price. The model was able to explain a significant portion of the variation in house prices and provided reasonably accurate predictions.

Recommendation:

Real estate businesses should focus on properties with larger areas, better facilities, and additional amenities such as air conditioning and parking. These features contribute significantly to higher property values and increased customer demand.

Conclusion:

This project successfully demonstrated the use of data preprocessing, visualization, and machine learning techniques for house price prediction.